

RECOMMENDATION TABLES

AREA OF USE	CROP/ SITUATION	TARGET WEEDS/USAGE	WEED INFESTATION	APPLICATION RATE L/HA	WATER VOLUME	APPLICATION TIMING AND GUIDANCE
ALL EDIBLE AND NON-EDIBLE CROPS (DESTRUCTION, BEFORE SOWING/PLANTING)	Vegetation management	-	Annual weeds	1.2	80-250 L/ha* or hand-held equipment section	*Rotary atomisers may be used at a water volume of 10-40 L/ha. Ensure droplet diameter falls within the range 200-300 microns Do not use in or alongside hedgerows Do not use under polythene or glass
			Perennial grasses	3.2		
			Perennial broad-leaved weeds	4.0		
GRASSLAND - DESTRUCTION	GRASS	Short rotation Ryegrass, longer leys and permanent pasture	Short rotation Ryegrass with annual weeds	2.4	150-250 L/ha*	Treat EITHER before grazing/mowing in June-Oct, when growth is 30-60 cm, not dense and lacking mature seeds, OR regrowth after grazing/mowing. Select the application rate which controls/destroys the least susceptible weed and grass species present in the sward. Grass may be conserved or grazed by cattle, dairy cows or sheep 5+ days after spraying. REMOVE POISONOUS PLANTS BEFORE GRAZING/MOWING. If Ragwort is present, the guidance in the DIRECTIONS FOR USE must be followed. ONLY direct drill grass and clover EITHER into 1-2 year leys without mat, 5+ days after spraying, OR long leys with some mat, in the spring following autumn application.
			Leys 2-4 years old with perennial grass weeds	3.2		
			Long leys 4-7 years old with perennial broad-leaved weeds	4.0		
			Permanent pasture See Weed Table below	4.8		

APPLICATION RATE FOR GRASSLAND DESTRUCTION							
2.4 L/HA		3.2 L/HA		4.0 L/HA		4.8 L/HA	
Annual Meadow-grass	Meadow Fescue	Black-bent	Creeping	Bracken**	Red Clover	Common Ragwort	<i>Nardus</i> (Mat grass)
Common Chickweed	Meadow Foxtail	Broad-leaved Dock	Soft-grass	Common Sorrel	Sedges	Hard Rush	Red Fescue
Common Mouse-ear	Rough Meadow-grass	Cock's-foot	Curled Dock	Common Nettle	Sheep's Sorrel	Heath Rush	White Clover*
Dock Seedlings	Speedwell species	Common Bent	Perennial Rye-grass	Creeping Buttercup*	Soft Rush	Jointed Rush	Yellow Rattle
Italian Rye-grass	Timothy	Common Couch	Plantains	Creeping Thistle	Spear Thistle	<i>Molinia</i> (Purple Moor-grass)	Sheep's Fescue
Mayweed species		Creeping Bent	Soft Brome		Tufted Hairgrass		
			Yorkshire Fog		Yarrow		
					Perennial Sow-thistle		

* White clover is best cut in June and sprayed one month later ** At full frond expansion

AREA OF USE	CROP/SITUATION	TARGET WEEDS/ USAGE	WEED INFESTATION	APPLICATION RATE L/HA	WATER VOLUME	APPLICATION TIMING AND GUIDANCE
NATURAL SURFACES NOT INTENDED TO BEAR VEGETATION, PERMEABLE SURFACES OVERLYING SOIL, RAILWAY BALLAST	Vegetation management	Including roadsides, paths and along fences	Annual weeds	1.2	Hydraulic sprayers (boom and knapsack) at water volumes 80-250 L/ha or Rotary atomisers* or hand-held equipment. See Mixing & Spraying section.	Do not use under polythene or glass.
			Perennial grasses and broad-leaved weeds	4.0		
HARD SURFACES (EXCLUDING RAILWAY BALLAST)	Vegetation management	including roadsides, paths, concrete and alongside walls	Annual weeds	1.2	Hydraulic sprayers (boom and knapsack) at water volumes 80-250 L/ha or Rotary atomisers* or hand-held equipment. See Mixing & Spraying section.	Apply this product carefully. Ensure spraying takes place only when weeds are actively growing (normally March to October) and is confined only to visible weeds including those in the 30 cm swath covering the kerb edge and road gully – do not overspray drains
			Perennial grasses and broad-leaved weeds	4.0		
AMENITY VEGETATION	Vegetation management	Ornamental areas	Annual weeds	1.2	Hydraulic sprayers (boom and knapsack) at water volumes 80-400 L/ha or Rotary atomisers* at water volumes 10-40 L/ha or hand-held equipment. See Mixing & Spraying section.	Do not use under polythene or glass.
			Perennial grasses and broad-leaved weeds	4.0		

*Rotary atomisers may be used at a water volume of 10–40 L/ha. Ensure droplet diameter falls within the range 200-300 microns

Japanese Knotweed control

Japanese Knotweed is an invasive alien species reducing biodiversity in areas where it become established and propagating from tiny fragments of root, often spreading along watercourses. It is scheduled under the Wildlife and Countryside Act 1981 and all parts of the plant must be treated as Controlled Waste under the Environmental Protection Act 1990.

Roundup ProVantage XL can be used alone as part of an eradication programme or as part of an integrated programme in conjunction with soil disturbance or removal. Dormant rhizomes will not be controlled by Roundup ProVantage XL, but may be stimulated to grow by soil disturbance and then sprayed. It is particularly suitable for use near water. Sites must be monitored for at least three years and re-treated as necessary.

AREA OF USE	TARGET WEED	METHOD	APPLICATION RATE / WATER VOLUME	APPLICATION TIMING AND GUIDANCE
Amenity vegetation; Forestry; Natural surfaces not intended to bear vegetation, permeable surfaces overlying soil, hard surfaces; Enclosed waters, open waters, land immediately adjacent to aquatic area	Japanese Knotweed	Foliar application	4.0 L/ha Hydraulic sprayers 80-250 L/ha or hand-held equipment	For best results apply after the onset of flowering (usually August/September) but before dieback. Use specialist extending hand lances for stands 2-3 m tall. Good coverage is essential; spray the underside as well as the upper surface of the leaves. Or Only as part of an integrated programme, spray when stems are 1-1.5 m high (Usually at end of May) and repeat once re-growth reaches 1-1.5 m again later in same season or the following year. (More re-growth will occur from this timing.)
		Stem filling technique	10 ml of 16% solution per stem	See National Trust Methodology for full details*. Cut stems approx. 200 mm above base of cane & 40 mm above node. Rupture the central stem tissue with a screwdriver and use a spot gun to insert Roundup ProVantage XL into the hollow stem within 15 minutes of cutting.
		Stem injection technique	1.6 ml of neat solution per stem	Using specialist stem injection equipment, inject directly into individual stems.
		Hand-held weed wiper	1 part Roundup ProVantage XL to 2.75 parts water	Use where overall spraying is not desirable or target plants are small or unsuitable for stem filling, (< 8 mm) e.g. in re-treatment following foliar spraying.

AREA OF USE	CROP/SITUATION	TARGET WEEDS/USAGE	WEED INFESTATION	APPLICATION RATE L/HA	WATER VOLUME	APPLICATION TIMING AND GUIDANCE
ORCHARDS	APPLE, PEAR, PLUM, CHERRY, DAMSON	Perennial grasses, broad-leaved weeds	All levels of most species	4.0	200-400 L/ha	Spray AFTER autumn leaf-fall and BEFORE: Apples, pears - green cluster stage Stone fruit - white bud stage Treat root suckers in late spring only. Trees must have been established 2+ years. Avoid contact with tree 30+ cm above ground.
		Root suckers	-	4.0		
ENCLOSED WATERS, OPEN WATERS, LAND IMMEDIATELY ADJACENT TO AQUATIC AREAS	-	Emergded Weeds - Reeds, Rushes, Sedges, Grasses and Watercress	All levels/species	4.0	200-400 L/ha Optimum 250 L/ha or hand-held equipment (p.12)	Consult appropriate Environment Agency regional office before use. On water-lilies it is preferable to use a tractor or boat-mounted sprayer. During spraying do not exceed a pressure of 2.0 bars (30 p.s.i.). When using a tractor mounted sprayer do not exceed 8 kph (5 mph).
		Floating Weeds - White water-lily - Yellow water-lily	All levels	4.8	100-200 L/ha or hand- held equipment (p.12)	Use boat-mounted sprayers at slowest practical speed. Always apply against direction of any current. When disturbed by wash, WATER-LILIES may require re-treatment.

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AREA OF USE	CROP/SITUATION	TARGET WEEDS/USAGE	WEED INFESTATION	APPLICATION RATE L/HA	WATER VOLUME	APPLICATION TIMING AND GUIDANCE
GRASSLAND, PLANT FREE AREAS, ORNAMENTAL PLANTINGS, AMENITY VEGETATION	-	Individual weeds	All levels	1:2.75 dilution with water for wick-type weed wipers OR 1:1.5 dilution with water in hot, dry conditions. For 'new generation' wipers consult the manufacturer for guidance.		Weeds must be 10+ cm taller, and wiper 5+ cm higher, than desired vegetation. Contact Bayer Crop Science or your distributor for specific recommended weed wiper applicators. Treated POISONOUS WEEDS must be removed or allowed to completely degenerate before grazing or conserving. If Ragwort is present, the guidance in the 'DIRECTIONS FOR USE' must be followed.

Forestry weed control

Roundup ProVantage can be used for site preparation and for weed control in planted out trees.

AREA OF USE	TARGET WEEDS/USAGE	WEED INFESTATION	APPLICATION RATE L/HA.	WATER VOLUME	APPLICATION GUIDANCE
Forestry: - Pre-planting	Arable land, planting, replanting, & grassland areas	Arable weeds Grassland weeds	3.2 4.0	Hydraulic sprayers: 80-250 L/ha or rotary atomizers: 40 L/ha*	All tree species may be planted 7 days or more after treatment *Where rotary atomisers are used their droplet diameter must fall within the range 200-300µm.
Forestry: - Post-planting (directed) in conifers & broad-leaved trees	Clean-up around trees with knapsack applications.	Annual/perennial grasses and broad-leaves	3.2	Knapsack sprayers: 200-250 L/ha or Weed wiper mini: 1 part ROUNDUP PROVANTAGE XL to 2.75 parts water	It is ESSENTIAL to use a TREE GUARD for all applications made in the growing season. Treat bracken after frond tips are unfurled but before senescence. Treat heather late August to end September. All other woody weeds are treated June-August, before leaf senescence (but after new growth of crop has hardened).
		Woody weeds: Bracken/Beech Brush/Brambles Sycamore/Oak Hazel/Willow/Ash	2.4		
		Heather (peat soils) Heather (mineral soils)	3.2 4.8		
		Rhododendron (*)	8.0		Cut back and treat regrowth when at least 1 metre in height throughout the entire coppice.

(*) For improved control of Rhododendron add Mixture B (ADJ AO570) at a concentration of 2% final water volume to 6.0 L/ha of Roundup ProVantage XL. Application using the weed wiper is not suitable.

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AREA OF USE	TARGET WEEDS/USAGE	WEED INFESTATION	APPLICATION RATE L/HA.	WATER VOLUME	APPLICATION GUIDANCE
Forestry: - Post-planting (overall dormant season in certain conifers – conifer release)	Grass weeds - Lowland areas - Upland areas	Black Bent, Cock's-foot, Common Couch, Creeping Soft-grass, False Oat-grass, Fescues, Meadow-grasses, Other Bent species, Purple Moor-grass, Sweet Vernal-grass, Tufted Hair-grass, Wavy Hair-grass, Wood Small-reed (Bush grass)	1.2 1.6	Hydraulic sprayers: 200-250 L/ha or Hand-held equipment - see 'Mixing and Spraying' section	Species safe to spray when fully dormant and leader growth has hardened: Corsican, Lodgepole and Scots Pines, Norway Spruce, Sitka Spruce, Lawson Cypress, Western Red Cedar. Douglas Fir and Noble Fir – safe to spray when fully dormant and leader growth has hardened but NOT in spring. If overall application takes place after the optimum timing weed control may be reduced. It is advisable to spray a limited area of forest to test crop safety under local conditions before widespread overall application in subsequent years. These recommended application rates refer to forestry usage only. Inadequate control may result if used in other areas. <u>Caution:</u> The timing of hardening of leader growth varies considerably between locations and between seasons. It may occur as early as the end of July or be delayed to October or later. To avoid damage to Lammas growth, sprays should be directed away from leaders.
Forestry: - Stump application for chemical thinning	Deciduous trees Coniferous trees	All species All species	8% solution of Roundup ProVantage XL in water 16% solution of Roundup ProVantage XL in water		Apply the solution to saturate the rim of the newly cut surface, with a suitable adapted clearing saw, spot gun or paintbrush. Treat as soon as possible after felling, in the period November to March/April. Do not apply in the period of active sap flow in the spring/early summer. Do not cut trenches or drill holes and fill with the solution or use undiluted product. Note: for ease of identification of treated areas a suitable, commercially available, water-soluble dye may be added to the prepared spray solution.
Forestry: - Chemical thinning by injection of tree stems	Coniferous and deciduous species	-	1.6 ml neat Roundup ProVantage XL per cut per 10 cm diameter (or less) tree		Use a hatchet to cut one notch in trees up to 10 cm diameter and apply 1.6 ml of the solution to each cut. Use two or three notches in trees over 10 cm diameter. Do not treat in the period of active sap flow in the spring/ early summer.

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